

Ujjwal Omar

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EDUCATION

Degree	Institute	Year	Grade
B.Tech. (Electrical Engineering)	Harcourt Butler Technical University	2022–2026	7.74
Class XII (CBSE)	Army Public School, Kanpur	2021	94.4%
Class X (CBSE)	Air Force School, Kanpur	2019	94.3%

INTERNSHIP

Software Development Intern, SmartShala Edtech, Kanpur, UP June 2025

- Gained end-to-end software development experience by building and deploying full-stack applications, contributing across frontend, backend, database, and deployment workflows.
- Implemented REST APIs and backend services for AI-powered chatbot integration using LangChain.

PROJECTS

Static Site Generator for Blogging | *React, Express, Python, Pandoc, Vercel, LangChain* | Live: blog.dtbom.space ↗

- Engineered a full-stack system to transform and publish Markdown notes into a high-performance editorial website.
- Implemented a Python-driven validation engine and automated blog sync between user interface and website.
- Integrated LangChain-based analysis into the blog editor for content review and suggestions before publishing.

Block Diagram Reduction Tool | *JavaScript, FastAPI, Python, Pydantic* | Live: dtbom.space/digiblocker ↗

- Developed a full-stack web application for the stepwise reduction of block diagrams in control systems.
- Designed an interactive UI for constructing block diagrams with a structured step-by-step solution dashboard.
- Developed graph-based algorithms to apply topological reductions on block diagrams modelled as graphs.
- Implemented type validations with Pydantic, rate limiting middleware and async polling for solver status .

SKILLS

Programming: Python, JavaScript, SQL

Frontend: React, HTML, CSS, Tailwind

Backend: FastAPI, Pydantic, Node.js, Express.js, REST APIs, JWT Authentication

Database: MongoDB, Postgres, Prisma, MySQL

Tools: Git/GitHub, Vercel, Postman, LangChain

Core Concepts: DSA, OOP, DBMS, AI/ML, GenAI

Software Skills: API Design, Data Modelling, Backend Systems, Database Design, LLD, Deployment & Publishing

POSITIONS OF RESPONSIBILITY

- Led a team of 4 for Final Year Project, responsible for all programming and technical implementation.
- Led a team of 4 in Smart India Hackathon, 2024, pre-finalist, FPGA-based Gunshot Detector System.
- Contributed to documentation and reporting for Resonance 2025, the department's annual technical fest.

EXTRACURRICULAR AND ACHIEVEMENTS

- **HackerRank Certifications:** Python ↗ and Solved 100+ Problems in Leetcode and HackerRank.
- Member of Association of Electrical Engineers (AEE), HBTU
- Participated and ranked 4th in Simulink Competition organized by AEE in Resonance 2025.
- Numerical Techniques & Tools Workshop, organised by Mathematics Club, IIT Madras ↗
- Participated in HPE Codewars 2021 organized by Hewlett Packard Enterprise. ↗